Joschka Wiegleb

Postdoctoral Researcher and Co-Founder
Department of Environmental and Life Sciences
University of Karlstad, Sweden



Basics

Nationality: German
Date of birth: 12/05/1993
Married, three children

Contact

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h-index: 6
i10-index: 6
Citations: 186
Google Scholar
ResearchGate

Languages

German – native
English – proficient

Research experience

Since 09/2024 **PostDoc,** Department of Environmental and Life Sciences, University of Karlstad, Sweden, Research Project: Machine learning to improve eel passage design, Supervisor: Prof. Olle Calles

02/2023 – 08/2024 **PostDoc**, International Centre for Ecohydraulic Research (ICER), University of Southampton, United Kingdom, Research Project: Towards understanding fish swimming patterns in critical flow (Funded by the SNSF: 214358), Supervisor: Prof. Paul Kemp

01/2022 – 01/2023 **PostDoc**, Program MGU, University of Basel, Switzerland, Supervisor: Prof. Patricia Burkhardt-Holm

Innovation and Technology Transfer

Since 06/2026 **Co-Founder** and **Board Member** of ConnFlume Technologies AB, Karlstad, Sweden. University spin-off translating basic hydrodynamics and fish behaviour into commercial applications to improve fish passage through culverts.

Education

12/2017 – 01/2022 **PhD** in Environmental Science, Dissertation title: 'Hydraulic burden and swimming behaviour of benthic fish in a vertical slot fish pass', University of Basel, Switzerland, defended on 06/01/2022, Supervisor: Prof. Patricia Burkhardt-Holm

10/2015 – 09/2017 **M.Sc. Aquaculture**, thesis title: 'About the feeding ecology of the round goby', University of Rostock, Germany, Thuenen-Institute of Baltic Sea Fisheries, Supervisor: Dr. Daniel Oesterwind

10/2012 – 09/2015 **B.Sc. Biology**, thesis title: 'Construction of an aquatic system for incubation of cod (*Gadus morhua*) and lumpsucker (*Cyclopterus lumpus*) eggs with documentation of their development', University of Düsseldorf, Germany, Supervisor: Prof. Christopher Bridges

Practical experience

07/2017 – 08/2017 Research assistant, Thuenen-Institute of Baltic Sea Fisheries, Rostock, Germany

06/2017 – 07/2017	Voluntary work, Thuenen-Institute of Baltic Sea Fisheries, Rostock, Germany
01/2017 – 02/2017 08/2016 – 09/2016 (2 months)	Internship, Thuenen-Institute of Baltic Sea Fisheries, Rostock, Germany
06/2016 – 02/2017	Research assistant, Mecklenburg-Vorpommern Research Centre for Agriculture and Fisheries, Institute of Fisheries, Research Station Aquaculture, Born a. Darss, Germany
09/2015 – 10/2015	Internship, Mecklenburg-Vorpommern Research Centre for Agriculture and Fisheries, Institute of Fisheries, Research Station Aquaculture, Born a. Darss, Germany

Supervision of students

08/2022 – 05/2023	**Supervision** Master thesis of Lucas Bergmann, 'Microplastics pollution and animal swimming behaviour', Master of Sustainable Development, University of Basel, Switzerland
06/2021 – 03/2022	**Mentorship** Master thesis of Nandhakumar Govindasamy, 'What is the physical benefit of staying close to the ground in flowing water in benthic fish?', Master of Sustainable Development, University of Basel, Switzerland

Teaching

SpS (Spring semester) 2022	Tools and Methods of Natural Sciences Research, 3 ECTS (Credit Points)
FS (Fall semester) 2019, FS 2020, FS 2021, FS 2022	Critical Reading Seminar, Master of Sustainable Development, University of Basel, 1 ECTS
02/2020	One-day workshop 'Fischwohl und Forschung' in 'Fachkurs Aquakultur' of the University of Applied Sciences Bern, Switzerland
FS 2018 – FS 2022	Organisation of the Life Science Kolloquium, Master of Sustainable Development, University of Basel, 1 ECTS

Review activity for scientific journals

06/2020	Article review for 'Journal of Aquacultural Engineering'
10/2020	Article review for 'Journal of Great Lakes Research'
08/2021	Article review for 'Ecology and Evolution'
11/2021	Article review for 'Ecology and Evolution'

Acquired third-party funds

06/2026	Verification grant (VFT) for Fish Rest application verification in dams (Innovationskontoret Fyrklöven, Grant No: 2026-0462)(110,000 SEK)

01/2026	Co-applicant for a Verification grant (VFT) to confirm IP protection by a U.S. provisional application (Innovationskontoret Fyrklöven, Grant No: 2025-0429)(31,250 SEK)
01/2026	Co-applicant for a Verification grant (VFT) for Verification (proof of concept) applications in culverts and fisways (Innovationskontoret Fyrklöven, Grant No: 2025-0415)(69,000 SEK)
11/2025	Co-applicant for a Verification grant (VFT) for a novelty search for the potentially patentable "Fish Rest Net" invention (Innovationskontoret Fyrklöven, Grant No: 2025-0408)(25,000 SEK)
12/2022	Postdoc.Mobility fellowship of the Swiss National Science Foundation (SNSF), amount: 190,139 CHF, project number: 214358, duration: 24 months
SpS 2021	Funding for scientific assistant support for one semester from the program 'Get on Track' of the University of Basel to support PhD students in parenthood (ca. 2,800 CHF)
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Prizes and Awards

08/2022	Second Prize 'Best Poster' at the International Congress of Ecology (INTECOL) in Geneva, Switzerland

Personal skills

R	statistics
Matlab	Data analysis and automation
Python	Data Management, Analysis and Deep Learning
Blender	3D Modelling and Video Processing
FreeCad	3D Modelling and Prototyping
3D printing	Prototyping
OpenFoam	Computational Fluid Dynamics
Basic electronics	Robotics, Prototyping

Continuing education

12/2025	Course: Teaching at University: Knowledge, learning and practical teaching (KLL-2510) (80h)
SpS 2021	Course: Konstruktive Konfliktlösung (30 h working hours), organizer: University of Basel, Switzerland
SpS 2021	Course: Boost Your Career: Fit for Own Third-Party Funding (30 h working hours), organizer: University of Basel, Switzerland
02/2021	

	Course: Python Crash Course (for Beginners) (10 h working hours), organizer: University of Basel, Switzerland
FS 2020	
FS 2020	Course: Meine erste Lehrveranstaltung unter Berücksichtigung der digitalen Lehre (30 h Arbeitsaufwand), organizer: University of Basel, Switzerland
FS 2020	Course: Writing for Publication in the Natural and Life Sciences (30 h working hours), organizer: University of Basel, Switzerland
05/2019	Course: Writing Workflow: Forming Your Next Research Article (30 h working hours), organizer: University of Basel, Switzerland
FS 2019	Course: Optimizing Research Data Management (14 h working hours), organizer: University of Basel, Switzerland
01/2019	Course: Rapid prototyping for measurement systems, automation, control, artificial intelligence, and virtual reality (60 h working hours), organizer: University of Basel, Switzerland
FS 2018	Course: Das Tierwohl (Bestandteil des Fachkurses Aquakultur) (8 h working hours), organizer: University of Applied Science Bern, Switzerland
12/2018	Course: Filmmaking for Scientists (30 h working hours), organizer: University of Basel, Switzerland
10/2018	LTK Moul 6E: Weighing of interests (6 h working hours), organizer: ETH Zürich, Switzerland
03/2018	Tierschutz-Kurs P: Arttypisches Verhalten: Haltung, Betreuung und Pflege von Fischen in der Praxis (6.5 h working hours), organizer: Ecotoxsolutions, Switzerland
	LTK Modul EGA: Introductory Course in Laboratory Animal Science (8 h working hours), organizer: ETH Zürich, Switzerland

Additional Qualifications

Basics of Laboratory Animal Science according to FELASA B

Exam for Ordinance on Aptitude of Instructors of the Chamber of Commerce and Industry, Rostock, achieved points: 89 / 94, grade: good

International Certificate for Operators of Pleasure Craft in Inland Waters

International Certificate for Operators of Pleasure Craft on the Waterways navigable by sea-going ships

Driver’s license, class B